(i) Let $u' = x^n \, \mathrm{d}x$, $v = \arctan x$

$\implies u = \dfrac{1}{n+1} \, x^{n+1}$, $v = \dfrac{1}{1+x^2} \, \mathrm{d}x$

$$\int uv' \mathrm{d}x = uv - \int u'v \mathrm{d}x$$

$$\begin{aligned} I_n &= \left[\frac{x^{n+1}}{n+1} \arctan x \right]_0^1 - \int_0^1 \frac{x^{n+1}}{n+1} \cdot \frac{1}{1+x^2} \, \mathrm{d}x \\ &= \frac{1}{n+1} \, \frac{\pi}{4} - \frac{1}{n+1} \int_0^1 \frac{x^{n+1}}{1+x^2} \, \mathrm{d}x \\ (n+1) \, I_n &= \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2} \, \mathrm{d}x \end{aligned}$$

When $n = 0$, $\quad I_0 = \dfrac{\pi}{4} - \displaystyle\int_0^1 \frac{x}{1+x^2} \, \mathrm{d}x = \frac{\pi}{4} - \frac{1}{2} \ln 2.$

(ii) From part (i):

$$(n+3) I_{n+2} = \frac{\pi}{4} - \int_0^1 \frac{x^{n+3}}{1+x^2} \, \mathrm{d}x, \qquad (n+1) I_n = \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2} \, \mathrm{d}x$$

$$\begin{aligned} (n+3) \, I_{n+2} + (n+1) \, I_n &= \frac{\pi}{2} - \int_0^1 \frac{x^{n+1}(x^2+1)}{1+x^2} \, \mathrm{d}x \\ &= \frac{\pi}{2} - \int_0^1 x^{n+1} \, \mathrm{d}x \\ &= \frac{\pi}{2} - \frac{1}{n+2} \end{aligned}$$

$$\boxed{(n+3) \, I_{n+2} + (n+1) \, I_n = \frac{\pi}{2} - \frac{1}{n+2}.}$$

When $n = 2$, $\quad 5I_4 + 3I_2 = \dfrac{\pi}{2} - \dfrac{1}{4}$

When $n = 0$, $\quad 3I_2 + I_0 = \dfrac{\pi}{2} - \dfrac{1}{2}$

$\implies 3I_2 = \dfrac{\pi}{2} - \dfrac{1}{2} - \left(\dfrac{\pi}{4} - \dfrac{\ln 2}{2} \right) = \dfrac{\pi}{4} - \dfrac{1}{2} + \dfrac{\ln 2}{2}$

$\implies 5I_4 = \dfrac{\pi}{2} - \dfrac{1}{4} - 3I_2 = \dfrac{\pi}{2} - \dfrac{1}{4} - \dfrac{\pi}{4} + \dfrac{1}{2} - \dfrac{\ln 2}{2} = \dfrac{\pi}{4} + \dfrac{1}{4} - \dfrac{\ln 2}{2}.$

$$\boxed{I_4 = \frac{\pi}{20} + \frac{1}{20} - \frac{\ln 2}{10}.}$$

(iii) $n = 1:$ LHS $= 5I_4$

$$\text{RHS} = A - \frac{1}{2}\left(-1 + \frac{1}{2}\right)$$
$$= A + \frac{1}{4}$$

$\because 5I_4 = \dfrac{\pi}{4} - \dfrac{1}{4} - \dfrac{\ln 2}{2}$

$\therefore$ In order to make LHS=RHS, $A = \dfrac{\pi}{4} - \dfrac{\ln 2}{2}$

Assume the equation is true for $n = k$:

$$(4k+1)\, I_{4k} = A - \frac{1}{2}\sum_{r=1}^{2k}(-1)^{r+1}\frac{1}{2r-1}$$

When $n = k + 1$:

Apply (ii) with $n = 4k$: $\quad (4k+3)\, I_{4k+2} + (4k+1)\, I_{4k} = \dfrac{\pi}{2} - \dfrac{1}{4k+2}$

$$(4k+3)\, I_{4k+2} = \frac{\pi}{2} - \frac{1}{4k+2} - (4k+1)\, I_{4k}$$
$$= \frac{\pi}{2} - \frac{1}{2(2k+1)} - A + \frac{1}{2}\sum_{r=1}^{2k}(-1)^r\frac{1}{r}$$
$$= \frac{\pi}{4} + \frac{\ln 2}{2} - \frac{1}{2(2k+1)} + \frac{1}{2}\sum_{r=1}^{2k}(-1)^r\frac{1}{r}$$

$$\because -\frac{1}{2(2k+1)} = \frac{1}{2}\cdot\frac{(-1)^{2k+1}}{2k+1}$$

$$\boxed{\therefore (4k+3)\, I_{4k+2} = \frac{\pi}{4} + \frac{\ln 2}{2} + \frac{1}{2}\sum_{r=1}^{2k+1}(-1)^r\frac{1}{r}}$$

Apply (ii) with $n = 4k + 2$: $\quad (4k+5)\, I_{4k+4} + (4k+3)\, I_{4k+2} = \dfrac{\pi}{2} - \dfrac{1}{4k+4}$

$$(4k+5)\, I_{4(k+1)} = \frac{\pi}{2} - \frac{1}{2(2k+2)} - \frac{\pi}{4} - \frac{\ln 2}{2} - \frac{1}{2}\sum_{r=1}^{2k+1}(-1)^r\frac{1}{r}$$
$$= \frac{\pi}{4} - \frac{\ln 2}{2} - \frac{1}{2(2k+2)} - \frac{1}{2}\sum_{r=1}^{2k+1}(-1)^r\frac{1}{r}$$

$\because (-1)^{2k+2} = 1 \qquad \therefore -\dfrac{1}{2(2k+2)} = -\dfrac{1}{2}\cdot\dfrac{(-1)^{2k+2}}{2k+2}$

$$(4k+5)\, I_{4(k+1)} = A - \frac{1}{2}\sum_{r=1}^{2(k+1)}(-1)^r\frac{1}{r}$$

By mathematical induction the result is true for all $n \geqslant 1$, with $\boxed{A = \frac{\pi}{4} - \frac{\ln 2}{2}}$.

This solution was submitted by **Jon**.

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